Matrices, Vectors & Complex Numbers

Section 1: Operations with Complex Numbers

1 What is the product of the complex numbers $(4 - 3i)$ and $(2 + 5i)$, where $i^2 = -1$?

a. $23 + 14i$

b. $7 + 14i$

c. $23 - 14i$

d. $8 - 15i$

2 What is the simplified form of the quotient $\frac{5 - i}{2 + i}$?

a. $\frac{9 - 7i}{5}$

b. $\frac{11 - 7i}{5}$

c. $3 - i$

d. $\frac{9 + 3i}{5}$

3 What is the value of i^{47}, where $i = \sqrt{-1}$?

a. 1

b. -1

c. i

d. $-i$

Section 2: Modulus & Graphing in the Complex Plane

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4 What is the modulus (absolute value) of the complex number $z = -5 + 12i$?

a. 7

b. 13

c. 17

d. $\sqrt{119}$

5 What is the distance in the complex plane between $z_1 = 3 - 2i$ and $z_2 = -1 + 4i$?

a. $2\sqrt{13}$

b. $\sqrt{20}$

c. 10

d. 5

6 If $z = 2 + 2i$, what is the argument θ of z in radians, where $0 \le \theta < 2\pi$?

a. $\frac{\pi}{6}$

b. $\frac{\pi}{4}$

c. $\frac{\pi}{3}$

d. $\frac{3\pi}{4}$

Section 3: Matrix Operations & Determinants

7 Given matrices $A = [[3, -1], [2, 4]]$ and $B = [[1, 5], [-2, 3]]$, what is the matrix $2A - B$?

a. $[[5, -7], [6, 5]]$

b. $[[5, 3], [2, 5]]$

c. $[[7, -7], [6, 11]]$

d. $[[5, -7], [2, 5]]$

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8 What is the determinant of the 2×2 matrix $M = [[4, 6], [-2, 3]]$?

a. 0

b. 12

c. 24

d. 36

9 For what value of k will the matrix $K = [[k, 6], [2, 3]]$ NOT have an inverse?

a. 1

b. 2

c. 3

d. 4

Section 4: Vector Operations & Dot Product

10 Vectors $\mathbf{u} = \langle 3, -4 \rangle$ and $\mathbf{v} = \langle -2, 5 \rangle$ are given. What are the components of $2\mathbf{u} + \mathbf{v}$?

a. $\langle 4, -3 \rangle$

b. $\langle 1, 1 \rangle$

c. $\langle 4, 3 \rangle$

d. $\langle -1, -3 \rangle$

11 What is the dot product of the vectors $\mathbf{a} = \langle 5, -2 \rangle$ and $\mathbf{b} = \langle 4, 7 \rangle$?

a. 6

b. 26

c. -6

d. 34

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12 What is the magnitude of the vector $\mathbf{w} = \langle -9, 12 \rangle$?

a. 13

b. 15

c. 17

d. 21

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